

ChargeFlow

AI-Powered EV Charging Grid Balancer & Infrastructure Optimization

Author: Antigravity AI Assistant

Status: Project Implementation Phase 1 Completed

Target Location: Coimbatore, Tamil Nadu, India

Date: June 2026

Frameworks: Spring Boot (REST Backend) + React Vite (TypeScript)

1. Executive Summary

ChargeFlow is an enterprise-grade smart EV routing and network optimization application designed to solve grid congestion and charging queuing delays. Rather than pointing drivers to the physically closest charging station, ChargeFlow models charging queues as an **Erlang-C queueing model**, combined with predictive analytics to route vehicles to stations that minimize *total expected charging delay* (travel time + expected waiting time + charging time) at the moment of arrival.

During this iteration, the infrastructure setup has been moved to center fully on the metropolitan region of **Coimbatore, Tamil Nadu, India**, establishing a localized network node seed database. Crucial UI bugs (layout overlap, click event validation handlers) and backend lazy-load database serialization errors have been successfully addressed, delivering a robust, compile-clean end-to-end flow.

2. System Architecture

The project consists of a high-performance decoupled architecture designed for real-time telemetry scaling:

- **Backend Service:** Java Spring Boot REST API backed by a MySQL database. Handles user authentication, token-based session validation, charging station configurations, active charging session locks, history logs, and runs the simulation engines.
- **Erlang-C Queue Prediction Engine:** Computes arrival rates, service rates, and utilization percentages to predict expected waiting times for arrival windows.
- **Dynamic Surge Pricing:** Automatically adjusts charging prices based on real-time station congestion and active queue lengths to balance demand via monetary incentives.
- **Frontend Dashboard:** Responsive React & TypeScript application styled with custom, modern glassmorphism CSS, integrated with Leaflet maps for live geolocation mapping and real-time chart stats.

3. Technical Achievements & Implementations

3.1 Geolocation Mapping Shift to Coimbatore

All charging stations have been re-centered within the geofenced boundaries of Coimbatore. Coordinates are seeded around major hubs (Gandhipuram, Peelamedu, Saravanampatti, RS Puram, and Singanallur). Distance metrics use the haversine formula to compute direct travel matrices.

3.2 Geolocation Live GPS Tracking

Configured the React frontend to leverage browser HTML5 Geolocation (`navigator.geolocation`) on mount. It estimates distances dynamically using the driver's real, live location instead of arbitrary placeholders, with manual coordinate override sliders available on the dashboard.

3.3 UI Badge Overlap Resolution

Resolved a visual overlap where the absolute positioned ★ Recommended Optimal badge covered the total delay time text in the routing list card. The badge was refactored to flow inline next to the station name header with responsive wrapping, maintaining sleek glassmorphic aesthetics.

3.4 Interactive Action Errors & Scroll Feedback

Fixed a bug where clicking 'Start Charging Session' failed silently without feedback. The driver dashboard now intercepts error triggers (e.g. no vehicle selected) and automatically scrolls the view smoothly to the top to bring the error banner into focus, additionally firing a browser alert to guarantee immediate feedback.

3.5 JPA Serialization Proxy Fix

Identified and resolved a Jackson JSON serialization error caused by Hibernate's Lazy Loading mechanism. The Vehicle.user relation was throwing a ByteBuddyInterceptor type definition exception when returning vehicle lists from the REST endpoints. Adding the @JsonIgnore decorator on the lazy relation successfully resolved the serialisation blocker.

4. Coimbatore Charging Network Seed Data

The following stations are seeded in the database and utilized by the Erlang-C prediction algorithms:

Station Name	Latitude	Longitude	Total Chargers	Base Price/kWh
Gandhipuram Grid Hub	11.0173	76.9691	8	\$0.32
Peelamedu Tech Node	11.0264	76.9961	12	\$0.45
Saravanampatti IT Park	11.0792	76.9997	6	\$0.28
RS Puram Fast Plaza	11.0116	76.9452	10	\$0.38
Singanallur Transport Hub	11.0028	77.0225	4	\$0.35

5. Default Users & Simulation Credentials

For testing and grid simulation control, the application seeds the following credential profiles:

Username	Password	Role	Seeded Vehicle
admin	admin123	ROLE_ADMIN (Grid Admin)	None
operator	operator123	ROLE_OPERATOR (Station Operator)	None
driver_kavin	driver123	ROLE_DRIVER (EV Driver)	Hyundai Ioniq 5
tesla_sophia	driver123	ROLE_DRIVER (EV Driver)	Tesla Model Y
volt_john	driver123	ROLE_DRIVER (EV Driver)	Hyundai Ioniq 5

6. Project Conclusion

With the completion of the Coimbatore grid seed and driver validation updates, ChargeFlow successfully meets all core operational goals. The Erlang-C prediction engine is fully functional, dynamic pricing logic correctly responds to real-time congestion scenarios, and the driver client interface renders seamlessly with no visual overlaps or silent errors. The platform is ready for active driver testing and scaling.